# Physics GCE A Level 2021 - Paper 1

Duration: 1h30 | Cameroon GCE Board

Multiple choice paper (0780 Physics 1, June 2021, Advanced Level, one and a half hours). Answer ALL 50 questions; all questions carry equal marks. Each question has FOUR suggested answers A, B, C and D; mark only one answer for each question (if you mark more than one you score zero). Mobile phones are NOT allowed in the examination room. The use of a calculator and the Formulae book is allowed.

**Constants used in the answers** (the paper itself prints no data sheet): acceleration of free fall $g = 9.81\ \mathrm{m\,s^{-2}}$ (question 24 gives the same result with $g = 10\ \mathrm{m\,s^{-2}}$); specific heat capacity of water $4200\ \mathrm{J\,kg^{-1}\,K^{-1}}$ (question 32).

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**Section I (thirty five questions), Questions 1-35.** Directions: Each of the questions or incomplete statements in this section is followed by four suggested answers. Select the best answer in each case.

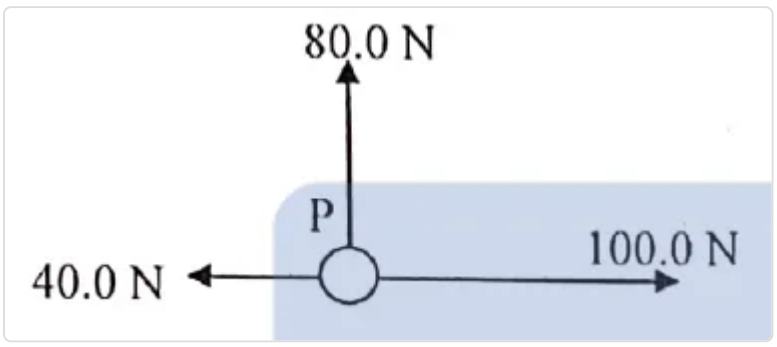
**Section II (ten questions), Multiple Selection, Questions 36-45.** Directions: for each of the questions or incomplete statements in this section ONE or TWO of the responses given is/are correct. Decide which of the response(s) is/are correct. Then choose: A - if 1 and 2 are correct; B - if 2 and 3 are correct; C - if 1 only is correct; D - if 3 only is correct. (Directions summarised: A = 1, 2 only; B = 2, 3 only; C = 1 only; D = 3 only.)

**Section III (five questions), Questions 46-50.** Directions: Each of the questions (46-50) has four sets of graphs A-D. Which of the graphs in each question best fits the relationship between $x$ and $y$?

**1.** Equality or equivalence in units for all terms in a physical equation makes the equation

a) homogeneous and correct
b) homogeneous only
c) physically correct only
d) homogeneous and possibly correct

**2.** The resultant force acting on object, P, in figure 1 is;




a) 100.0 N, North East of position of P
b) 80.0 N, due North of position of P
c) 60.0 N, due East of position of P
d) 40.0 N, South East of position of P

**3.** A system which can attain resonance

*Options as printed:* A) oscillates with large amplitude B) oscillates at its natural frequency C) encounters very little damping D) performs forced oscillations

**4.** Which of the following is correct for a body giving out heat?

a) Its temperature always drops with time
b) Its temperature could be constant
c) It cannot be in thermal equilibrium with any other body
d) It has a lower heat capacity than its surroundings

**5.** Which of the following is true for a body moving in a circular path at constant speed;

a) The acceleration is perpendicular to the velocity
b) The acceleration is constant with time
c) The acceleration is always directed opposite to the displacement
d) The centripetal force is provided by the weight of the object

**6.** When a beam of white light is incident on a diffraction grating, red light is deviated most because in the visible spectrum red light has the;

a) highest frequency
b) longest wavelength
c) highest intensity
d) lowest speed in air

**7.** The displacement of a particle performing simple harmonic motion with angular frequency, $\omega$, is given by $x = A\cos(\omega t - \varphi)$; where $t$ is the time and $A$ the amplitude. The total distance covered by the particle in two complete cycles is

a) 8A
b) 4A
c) A
d) 2A

**8.** If a car initially moving with a velocity of $4\ \mathrm{m\,s^{-1}}$ accelerates steadily at $2\ \mathrm{m\,s^{-2}}$ for $4$ seconds,

a) It covers a distance of 16 m
b) It attains a velocity of 8 m/s
c) Its velocity-time graph is a straight line with non-zero slope.
d) The frictional resistance to the motion is negligible.

**9.** According to Coulomb's law, the force between two point charged objects is doubled if the

a) charge on each object is doubled
b) distance between them is halved
c) charges are brought closer
d) product of the charges is doubled.

**10.** In an electric field of strength $E$, a charge $Q$ is accelerated through a distance $y$, the work done by the electric force is

a) QEy
b) QE/y
c) Qy/E
d) QE

**11.** Images of some personalities and messages are sometimes attached to inflated balloons and the systems are observed to rise in air when released because;

a) the air inside the balloons is always warmer than the air outside
b) up thrust on the inflated balloons is quite significant
c) inflated balloons have very low densities
d) inflated balloons have negligible weights

**12.** Figure 2 shows a low power lamp connected in series with a cell, V, a capacitor, C, and a switch, S, which is initially open.

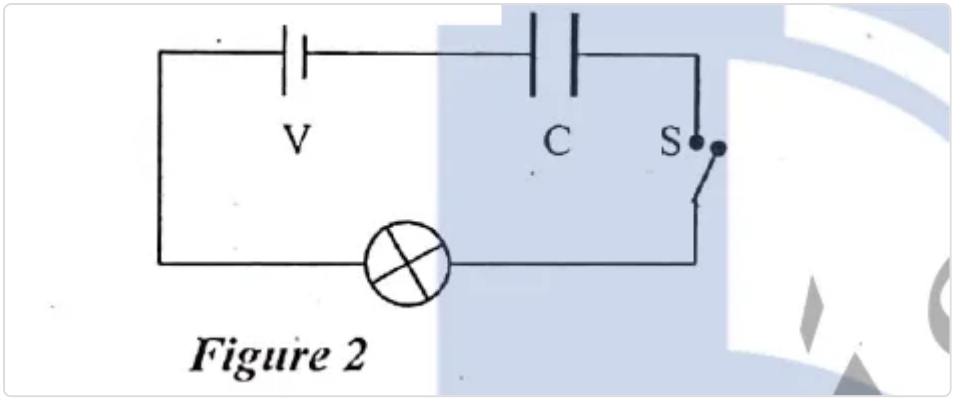
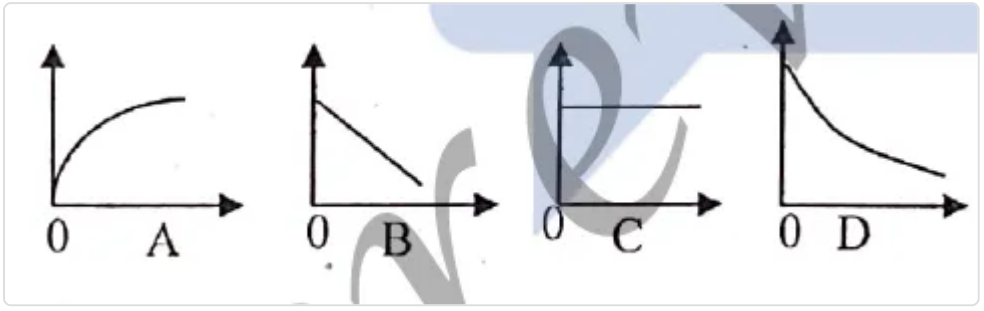


*Figure 2*

Which of the graphs represents the variation of the brightness of the lamp with time after the switch is closed assuming that the capacitor was initially uncharged.




a) A
b) B
c) C
d) D

**13.** A beam of light of wavelength $\lambda$ ejects photoelectrons from the surface of a clean metal plate. A less intense beam of light of the same wavelength would emit photoelectrons

a) equal in number and with same KE
b) less in number but with same KE.
c) less in number and with less KE
d) equal in number and with less KE

**14.** Figure 3 shows a potentiometer used to measure the emf, $E$, of a cell. A balanced point may NOT be obtained if

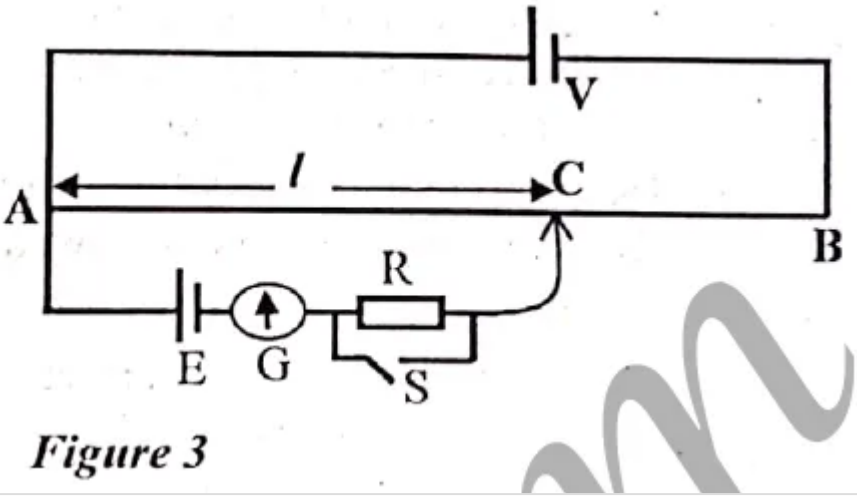


*Figure 3*

a) $E > V_{AC}$
b) $E = V_{AC}$
c) $E > V_{AB}$
d) $E < V_{AC}$

**15.** The activity of a radioactive substance reduces to $68\%$ of the original value in $2.82$ hours. The half-life of the substance in hours is

a) 5.07
b) 6.82
c) 4.30
d) 3.51

**16.** Photons of light enter a glass block after travelling through a vacuum. The energy of the photons on entering the glass block

a) increases because wavelength decreases
b) decreases because speed decreases
c) stays the same because speed and wavelength do not change.
d) stays the same because frequency does not change.

**17.** Laser beams are mostly very intense because

a) of population inversion
b) all electron transitions from the excited to the ground state are identical
c) a large number of electron transitions occur simultaneously
d) electrons emit energy as they fall from excited states to the ground state.

**18.** A boy adds $5.00\ \text{J}$ of energy to a girl on a swing by giving a push periodically each time she is at the same position of swing. After doing this continuously for $10$ minutes, a total of $2.40\ \text{kJ}$ of energy has been added to her on the swing. Her frequency of swing in Hz is

a) 0.34
b) 0.80
c) 0.57
d) 0.28

**19.** In Ndu Tea Estate, powder tea falls from the grinding machine at a rate of $0.80\ \text{kg s}^{-1}$ onto a conveyor belt moving horizontally at $2.50\ \text{m s}^{-1}$. Determine the force in newtons needed to keep the conveyor belt moving at this constant speed.

a) 2.00
b) 3.10
c) 0.32
d) 3.30

**20.** In the absence of air resistance, an object projected vertically upwards with speed, $v$ returns to the same level with speed, $v$ . This is an aspect of

a) the law of conservation of linear momentum
b) Newton's second law
c) Newton's third law
d) the law of conservation of energy

**21.** A car moving towards a police post at $40\ \mathrm{m\,s^{-1}}$ sounds its horn emitting sound of frequency $380\ \mathrm{Hz}$. If the speed of sound in air is $340\ \mathrm{m\,s^{-1}}$, the wavelength in metres of the horn as determined by the radar at the police post will be

a) 1.12
b) 0.79
c) 1.00
d) 0.89

**22.** A beam of protons used in an experiment is referred to as being a $20\ \mathrm{MeV}$ beam. The MeV is the unit of

a) charge
b) potential difference
c) energy
d) electric field intensity

**23.** If the switch, K, in figure 4 is closed,

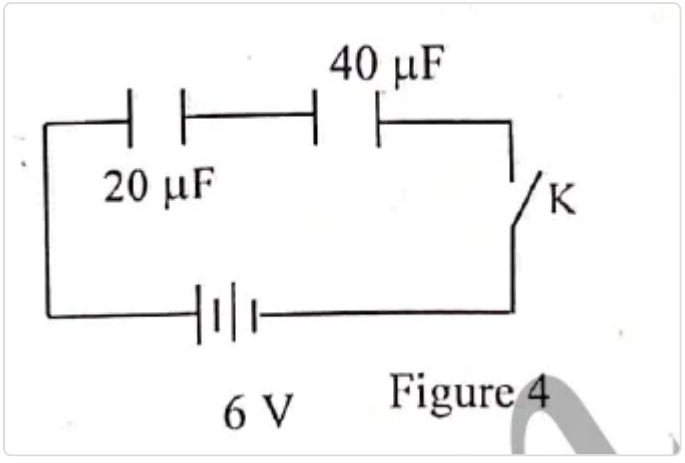
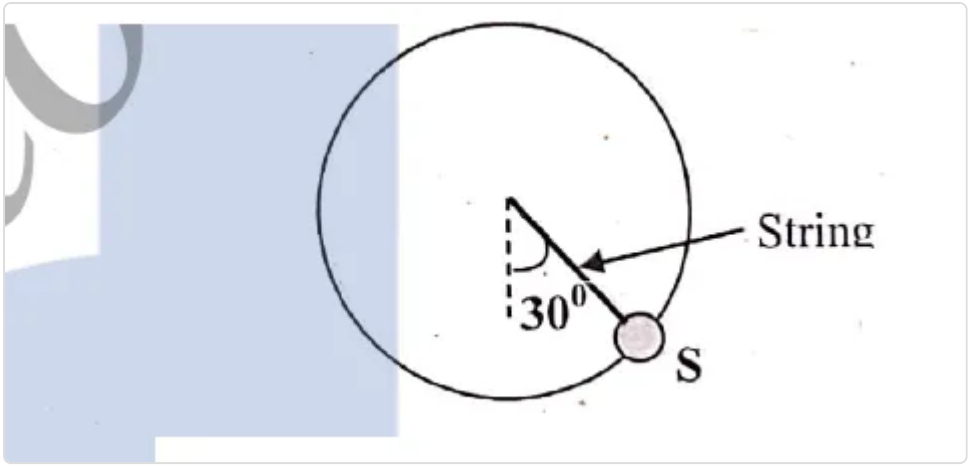


Figure 4

a) The effective capacitance of the circuit is 60 µF.
b) The p.d across the 40 µF capacitor is 4 V
c) The p.d across the 20 µF capacitor will be double the p.d across the 40 µF capacitor.
d) The capacitors will each store the same amount of energy.

**24.** Figure 5 shows a stone of mass $50.0\ \mathrm{g}$ tied to a $1.50\ \mathrm{m}$ long string and whirled in a vertical circle at the point where the string makes an angle of $30^{\circ}$ with the vertical.




If the stone makes $20$ revolutions in $5.00$ s. The tension in newtons on the string at this point is

a) 47.8
b) 46.7
c) 45.4
d) 48.0

**25.** If an object is placed close to each of a thin convex and a thin concave lens perpendicularly to the principal axis such that the distance from the lens is smaller than the focal length for each case; the image formed by the

a) convex lens may either be real or virtual while the image formed by the concave lens will be virtual only
b) convex lens will be real and inverted while the image formed by the concave lens will be virtual and diminished
c) concave lens will be real and diminished while the image formed by the convex lens will be virtual and larger than the object
d) concave lens will be virtual and diminished while the image formed by the convex lens will be virtual and larger than the object.

**26.** A parallel beam of monochromatic light of wavelength $520$ nm is incident on a diffraction grating. The grating has $400$ lines per millimetre. The maximum order of diffraction that can be observed is

a) 3
b) 5
c) 4
d) 6

**27.** The graph in figure 6 is obtained from an experiment to determine the temperature coefficient of resistance, $\alpha$, of a metal wire.

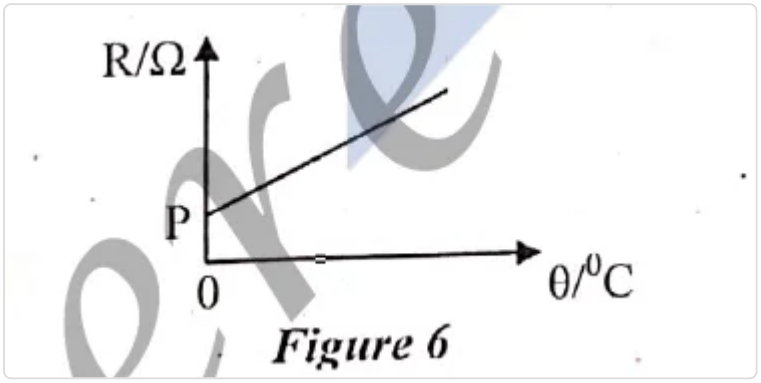


*Figure 6*

If the slope of the graph is S, then, $\alpha$ is given by

a) P/S
b) S/P
c) PS
d) 1/PS

**28.** In a heat engine, heat energy, $Q_H$ is absorbed by the engine to do mechanical work while some of the heat, $Q_C$ is not available for conversion. The efficiency of the engine is given by

a) $\left(1 - \frac{Q_C}{Q_H}\right) \times 100\%$
b) $\left(\frac{Q_C}{Q_H}\right) \times 100\%$
c) $\left(1 - \frac{Q_H}{Q_C}\right) \times 100\%$
d) $\left(\frac{Q_H}{Q_C}\right) \times 100\%$

**29.** A step down transformer with a turns ratio of $5:4$ and an efficiency of $88\%$ is used to run a motor. If the mains a.c. is $5.0\ \mathrm{A}$, the current drawn by the motor will be

a) 3.5
b) 8.8
c) 1.8
d) 5.5

**30.** The critical angle for a ray of light moving through an optical fibre from a core of refractive index $1.54$ to a cladding of refractive index $1.48$ is approximately

a) $80^\circ$
b) $43^\circ$
c) $74^\circ$
d) $40^\circ$

**31.** In atomic physics the binding energy of an atomic nucleus is the

a) energy used in forming the nucleus from constituent nucleons
b) Energy needed to completely separate the nucleons
c) energy equivalent of the mass of the nucleus
d) difference between the total mass of the nucleons and that of the entire nucleus.

**32.** Running water is used to remove excess heat at a rate of $120\ \mathrm{J\,s^{-1}}$ from a vacuum pump. At what rate in $\mathrm{kg\,s^{-1}}$ does the water flow if it enters the pump at $20\ ^\circ\mathrm{C}$ and leaves at $22\ ^\circ\mathrm{C}$?

a) 0.01
b) 0.08
c) 0.14
d) 0.28

**33.** A force of $0.60\ \mathrm{kN}$ is applied to a wire of diameter $0.002\ \mathrm{m}$ to cut it without the wire developing a 'neck'. Find the breaking stress in Pa on the wire.

a) $3.8 \times 10^5$
b) $9.6 \times 10^7$

c) $4.8 \times 10^7$
d) $1.9 \times 10^8$

**34.** A newly discovered planet has a mass equal to one eighth the mass of the earth and a radius equal to half of the earth's radius. If the gravitational field strength on the earth's surface is $g$, the gravitational field strength on the new planet's surface is

a) g/16
b) g/4
c) g
d) g/2

**35.** $^{214}_{82}\mathrm{Pb}$ decays by a series of transformations emitting $2$ alpha particles and $4$ beta particles to a final stable product. Which of the following could be the symbol for the stable nucleus formed?

a) $^{214}_{83}\mathrm{Bi}$
b) $^{214}_{84}\mathrm{Po}$
c) $^{206}_{82}\mathrm{Pb}$
d) $^{208}_{83}\mathrm{Pb}$

**36.** Figure 7 below shows a coil connected in series with a resistor and a battery

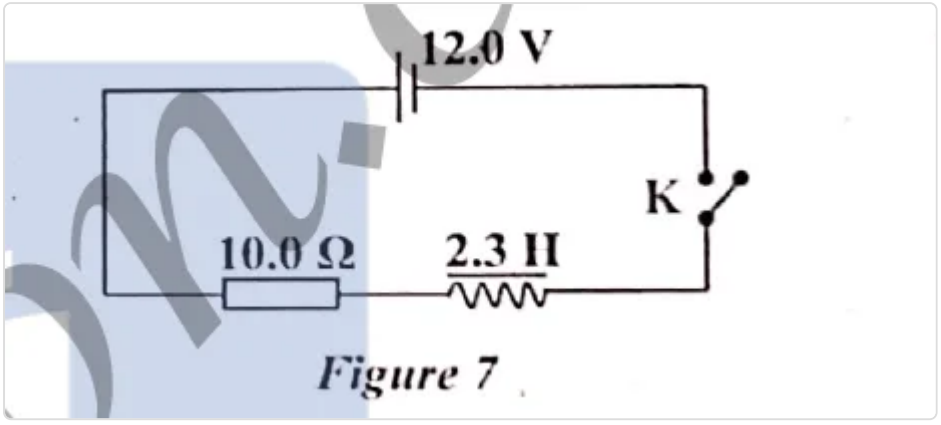


*Figure 7*

When K is closed?

1. The initial increase in current is $5.2\ \mathrm{A\,s^{-1}}$
2. The maximum current in L is $1.2\ \mathrm{A}$
3. The back emf across the coil drops with time

*Options as printed:* A) 1 and 2 are correct B) 2 and 3 are correct C) 1 only is correct D) 3 only is correct

**37.** The slope of a velocity - time graph for a car gives

1. its rate of change of momentum with time
2. its acceleration
3. the resultant force per unit mass acting on it.

a) 1 and 2 are correct
b) 2 and 3 are correct
c) 1 only is correct
d) 3 only is correct

**38.** A small magnetic needle in a compass is at rest on a horizontal table top in a science laboratory. Which of the following may cause deflection of the needle?

1. A straight wire carrying current
2. A collimated beam of beta particles
3. An intense LASER beam of monochromatic light

a) 1 and 2 are correct
b) 2 and 3 are correct
c) 1 only is correct
d) 3 only is correct

**39.** If the oscillations accompanying the propagation of a wave can be restricted to a single plane then it can be deduced that the wave could be;

1. transverse
2. electromagnetic
3. longitudinal

a) 1 and 2 are correct
b) 2 and 3 are correct
c) 1 only is correct
d) 3 only is correct

**40.** The resistance of a piece of copper wire varies with

1. resistivity
2. thickness
3. length

*Options as printed:* A) 1 and 2 are correct B) 2 and 3 are correct C) 1 only is correct D) 3 only is correct

**41.** Which of the following is/are conservative forces?

1. gravitational force
2. tension
3. friction

a) 1 and 2 are correct
b) 2 and 3 are correct
c) 1 only is correct
d) 3 only is correct

**42.** Two identical copper wires A and B carrying currents of magnitudes $2.5\ \mathrm{A}$ and $3.0\ \mathrm{A}$ respectively in the same direction are placed on a table $2.0\ \mathrm{cm}$ apart. Which of the following is/are true?

1. Force of B on A is greater than force of A on B
2. The force between them will be attractive.
3. The magnetic flux of A through B is less than that of B through A

a) 1 and 2 are correct
b) 2 and 3 are correct
c) 1 only is correct
d) 3 only is correct

**43.** The potential energy between a pair of molecules of a substance can be increased by

1. bringing them closer
2. cooling the substance
3. moving them apart

*Options as printed:* A) 1 and 2 are correct B) 2 and 3 are correct C) 1 only is correct D) 3 only is correct

**44.** Nuclear fusion could be better than nuclear fission in harnessing nuclear energy and converting it to useful forms because

1. fission occurs very slowly
2. fusion yields more energy than fission for equal masses of fuel
3. fuel for fusion is easily obtainable

a) 1 and 2 are correct
b) 2 and 3 are correct
c) 1 only is correct
d) 3 only is correct

**45.** Figure 8 shows a bridge circuit of 4 resistors.

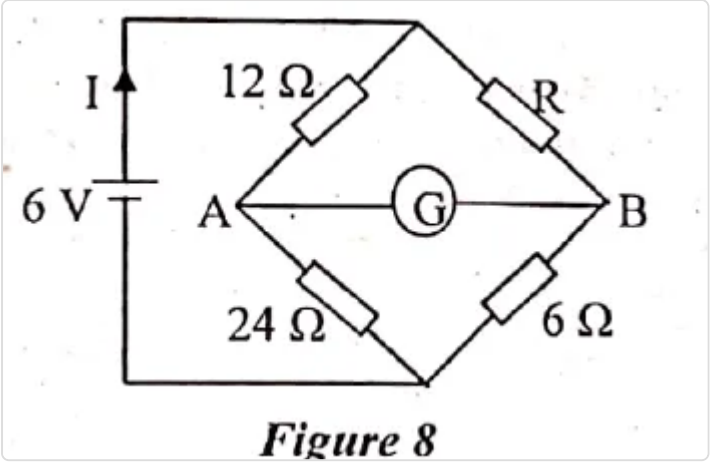
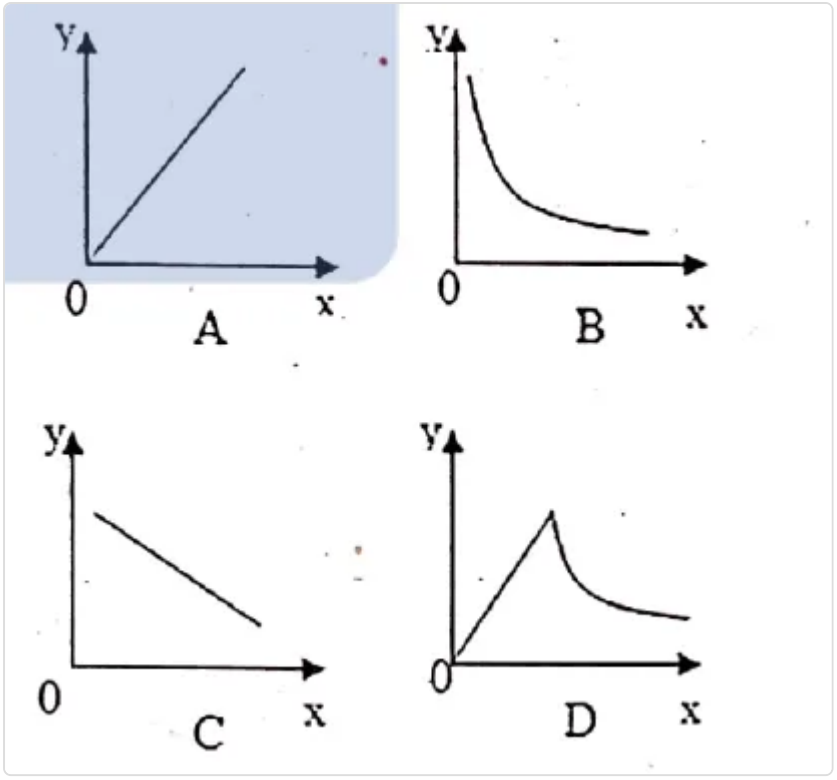


*Figure 8*

1. P.d across AB $= 6$ V
2. $I = 0.6$ A
3. $R = 3\ \Omega$

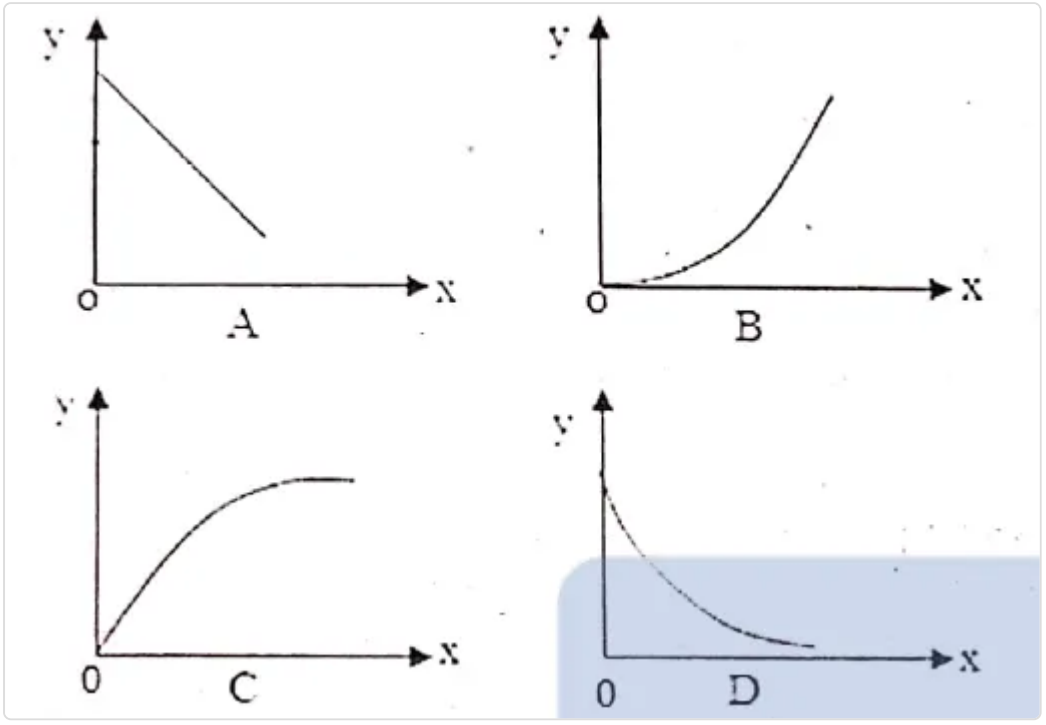
*Options as printed:* A) 1 and 2 are correct B) 2 and 3 are correct C) 1 only is correct D) 3 only is correct

**46.** The graphs below are for $y$ = the gravitational field strength at a point on the surface of a planet, and $x$ = the mean radius of the planet.



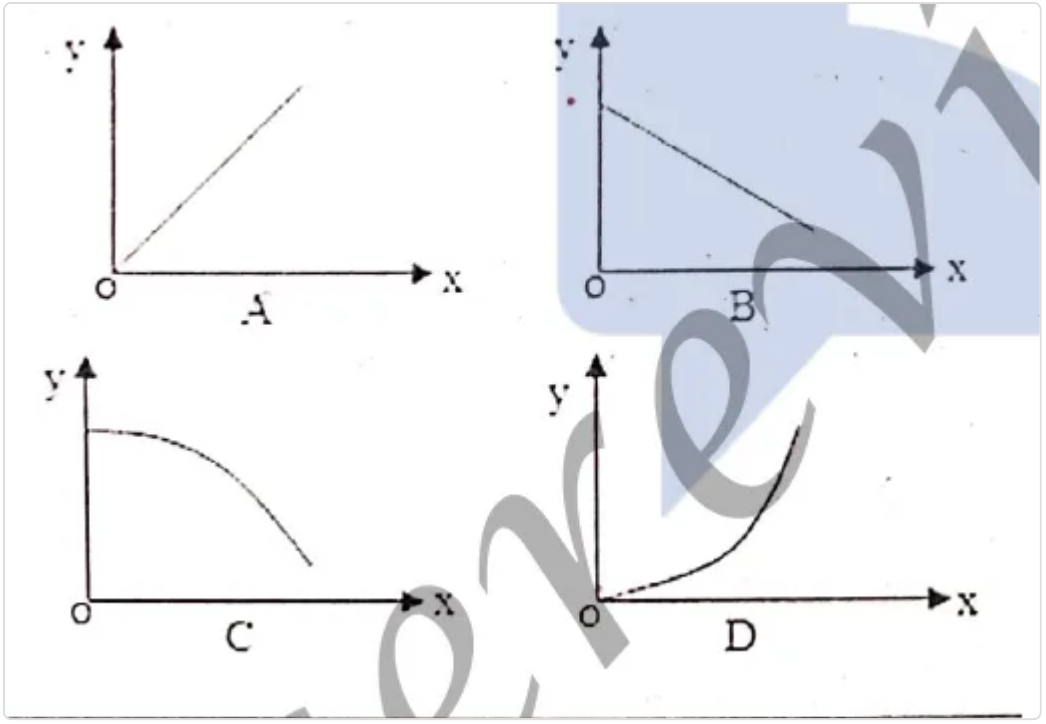

*Options as printed:* A) straight line through the origin, rising B) falling curve (like a hyperbola) that is steep at small $x$ and flattens C) straight line decreasing from a positive value on the $y$-axis D) straight line rising from the origin to a peak, then a falling curve

**47.** The graphs below are for $y$ = number of radioactive nuclei of a radioactive substance already decayed, and $x$ = time of decay.



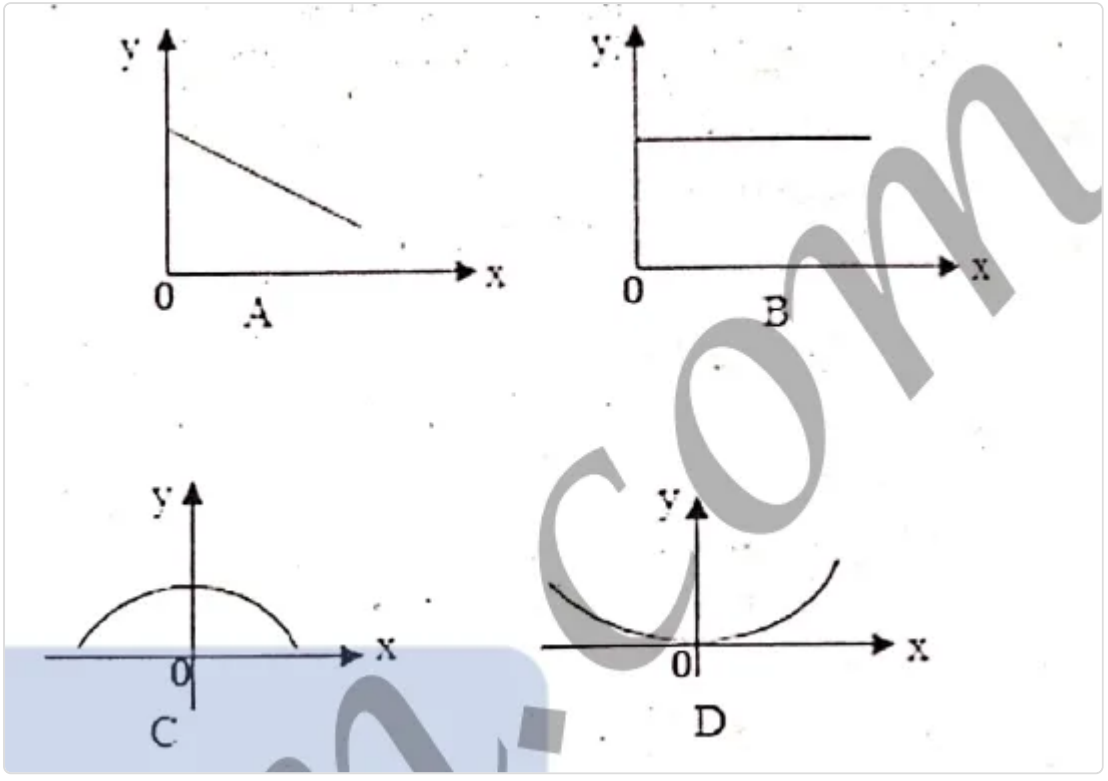

a) A
b) B
c) C
d) D

**48.** The graphs below are for $y$ = mean square speed of gas molecules, and $x$ = absolute temperature of the gas.



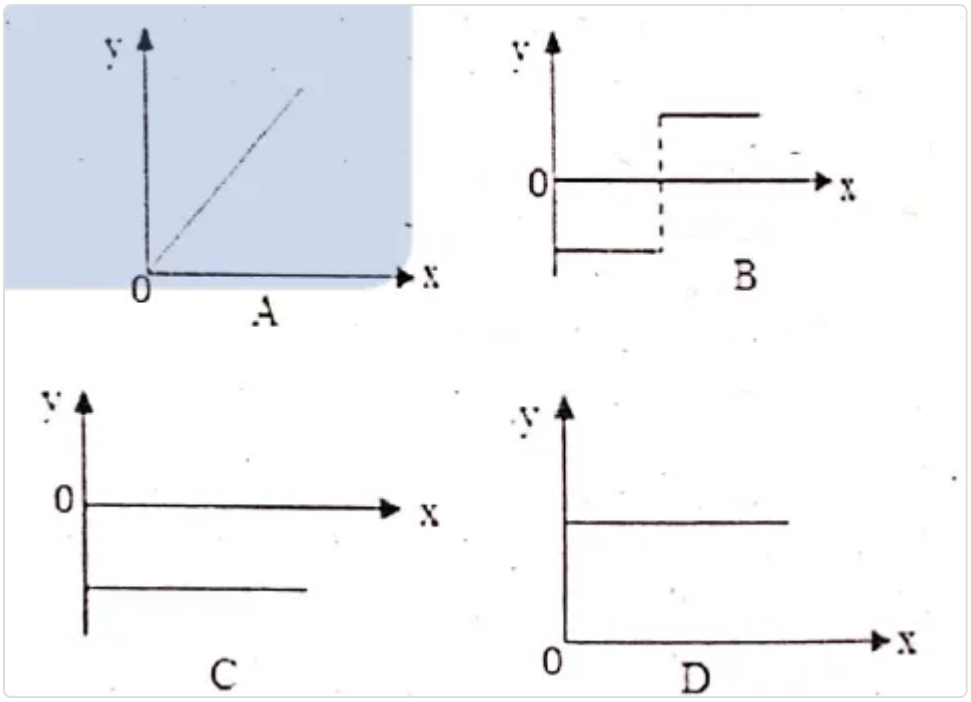

a) A
b) B
c) C
d) D

**49.** The graphs below are for $y$ = kinetic energy of a child on a swing, and $x$ = displacement from the rest position.




a) A
b) B
c) C
d) D

**50.** The graphs below are for $y$ = the acceleration of a ball kicked vertically upwards, and $x$ = time until the ball reaches maximum height.




a) A
b) B
c) C
d) D


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